EXPERIMENT 2

EXPLORATORY DATA ANALYSIS (EDA) – DATA IMPORT AND EXPORT

Aim

To import data from various sources such as CSV, Excel, SQL, and Web, perform basic inspection using Pandas, and export the processed data into different file formats.

Software Requirements

- Python 3.x
- Jupyter Notebook
- Pandas Library
- OpenPyXL Library
- SQLite Database
- Internet Connection (for Web Data)

Theory

Data Import and Export is the first step in Exploratory Data Analysis (EDA). Real-world data is stored in different formats such as:

- CSV Files
- Excel Files
- SQL Databases
- JSON Files
- Web APIs

Pandas provides powerful functions to load and save data efficiently.

Common Data Sources

Source	File Type
Spreadsheet	.xlsx
CSV	.csv
Database	SQL
Web API	JSON
Text File	.txt

Learning Outcomes

After completing this experiment, students will be able to:

✓ Import CSV files
✓ Import Excel files
✓ Read data from SQL databases
✓ Import data from Web URLs
✓ Export DataFrames to CSV and Excel
✓ Inspect imported datasets

Algorithm 1: Importing CSV File

In [11]:
```
import pandas as pd
import numpy as np
df=pd.read_csv(r"C:\Users\HDC0422042\Documents\231401013\healthcare-dataset-stroke-data.csv")
```

In [15]:
```
df.head()
```

Out[15]:

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

In [16]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 5110 non-null   int64
 1   gender             5110 non-null   object
 2   age                5110 non-null   float64
 3   hypertension       5110 non-null   int64
 4   heart_disease      5110 non-null   int64
 5   ever_married       5110 non-null   object
 6   work_type          5110 non-null   object
 7   Residence_type     5110 non-null   object
 8   avg_glucose_level  5110 non-null   float64
 9   bmi                4909 non-null   float64
 10  smoking_status     5110 non-null   object
 11  stroke             5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
```

In [17]:

```
df.describe()
```

Out[17]:

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	4909.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012	106.147677	28.893237	0.048728
std	21161.721625	22.612647	0.296607	0.226063	45.283560	7.854067	0.215320
min	67.000000	0.080000	0.000000	0.000000	55.120000	10.300000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000	77.245000	23.500000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000	91.885000	28.100000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000	114.090000	33.100000	0.000000
max	72940.000000	82.000000	1.000000	1.000000	271.740000	97.600000	1.000000

Algorithm 2: Importing Excel File

In [18]:

```
import pandas as pd
import numpy as np
df=pd.read_excel(r"C:\Users\HDC0422042\Documents\231401013\healthcare-dataset-stroke-data.xlsx")
```

In [20]:

```
df.head()
```

Out[20]:

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

In [16]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 5110 non-null   int64
 1   gender             5110 non-null   object
 2   age                5110 non-null   float64
 3   hypertension       5110 non-null   int64
 4   heart_disease      5110 non-null   int64
 5   ever_married       5110 non-null   object
 6   work_type          5110 non-null   object
 7   Residence_type     5110 non-null   object
 8   avg_glucose_level  5110 non-null   float64
 9   bmi                4909 non-null   float64
 10  smoking_status     5110 non-null   object
 11  stroke             5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
```

```
memory usage: 479.2+ KB
```

In [21]:

```
df.describe()
```

Out[21]:

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	4909.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012	106.147677	28.893237	0.048728
std	21161.721625	22.612647	0.296607	0.226063	45.283560	7.854067	0.215320
min	67.000000	0.080000	0.000000	0.000000	55.120000	10.300000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000	77.245000	23.500000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000	91.885000	28.100000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000	114.090000	33.100000	0.000000
max	72940.000000	82.000000	1.000000	1.000000	271.740000	97.600000	1.000000

Algorithm 3: Importing SQL Data

In [23]:

```
import pandas as pd
import sqlite3
df = pd.read_excel(r"C:\Users\HDC0422042\Documents\231401013\healthcare-dataset-stroke-data.xlsx")


conn = sqlite3.connect("stroke.db")


df.to_sql("stroke_data", conn, if_exists="replace", index=False)


query = "SELECT * FROM stroke_data"

sql_df = pd.read_sql(query, conn)

print(sql_df.head())

conn.close()
```

```
      id  gender   age  hypertension  heart_disease ever_married  \
0   9046    Male  67.0             0              1          Yes
1  51676  Female  61.0             0              0          Yes
2  31112    Male  80.0             0              1          Yes
3  60182  Female  49.0             0              0          Yes
4   1665  Female  79.0             1              0          Yes

       work_type Residence_type  avg_glucose_level   bmi   smoking_status  \
0        Private          Urban             228.69  36.6  formerly smoked
1  Self-employed          Rural             202.21   NaN     never smoked
2        Private          Rural             105.92  32.5     never smoked
3        Private          Urban             171.23  34.4           smokes
4  Self-employed          Rural             174.12  24.0     never smoked

   stroke
0       1
1       1
2       1
3       1
4       1
```

Algorithm 4: Importing Data From Web

```
[24]: import pandas as pd

url = "healthcare-dataset-stroke-data.csv"
df = pd.read_csv(url)

print(df.head())
```

```
      id  gender   age  hypertension  heart_disease ever_married  \
0   9046    Male  67.0             0              1          Yes
1  51676  Female  61.0             0              0          Yes
2  31112    Male  80.0             0              1          Yes
3  60182  Female  49.0             0              0          Yes
4   1665  Female  79.0             1              0          Yes

       work_type Residence_type  avg_glucose_level   bmi   smoking_status  \
0        Private          Urban             228.69  36.6  formerly smoked
1  Self-employed          Rural             202.21   NaN     never smoked
2        Private          Rural             105.92  32.5     never smoked
3        Private          Urban             171.23  34.4           smokes
4  Self-employed          Rural             174.12  24.0     never smoked

   stroke
0       1
1       1
2       1
3       1
4       1
```

Export to CSV

```
[1]: import pandas as pd

df = pd.read_csv(r"C:\Users\hp\Downloads\healthcare-dataset-stroke-data.csv")

df.to_csv("stroke_output.csv", index=False)

print("CSV File Exported Successfully")
```

```
CSV File Exported Successfully
```

Export to Excel

```
[4]: import pandas as pd

df = pd.read_csv(r"C:\Users\hp\Downloads\healthcare-dataset-stroke-data.csv")

df.to_excel("stroke_output.xlsx", index=False)

print("Excel File Exported Successfully")
```

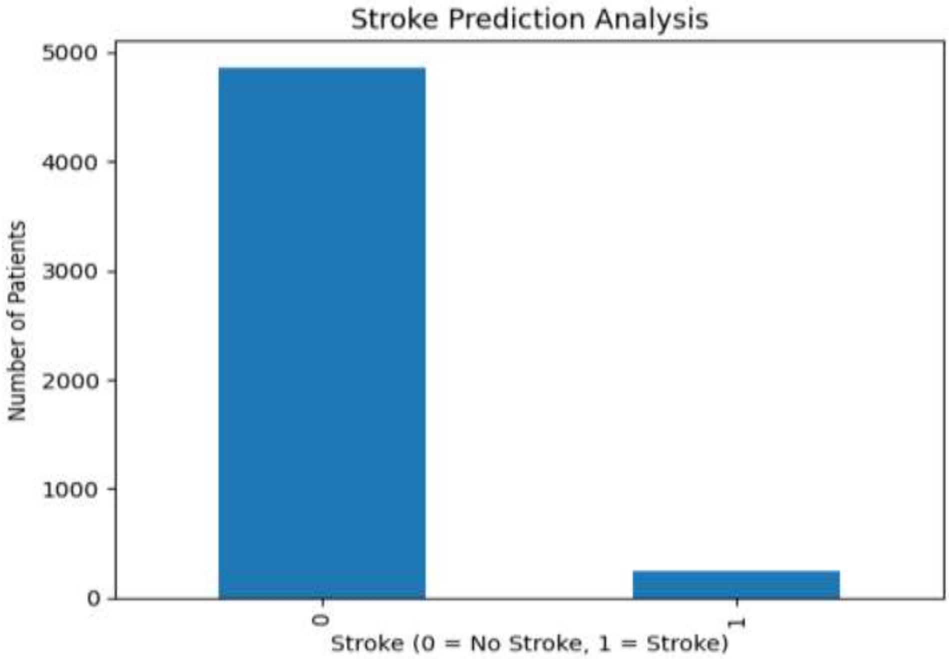
```
Excel File Exported Successfully
```

Export to JSON

```
5]: import pandas as pd

    df = pd.read_csv(r"C:\Users\hp\Downloads\healthcare-dataset-stroke-data.csv")

    df.to_json("stroke_output.json")

    print("JSON File Exported Successfully")

    JSON File Exported Successfully
```

Visualization of Imported Data

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\hp\Downloads\healthcare-dataset-stroke-data.csv")

stroke_counts = df["stroke"].value_counts()
print(stroke_counts)

stroke_counts.plot(kind="bar")

plt.title("Stroke Prediction Analysis")
plt.xlabel("Stroke (0 = No Stroke, 1 = Stroke)")
plt.ylabel("Number of Patients")

plt.show()

stroke
0    4861
1     249
Name: count, dtype: int64
```



Result

Thus, data was successfully imported from CSV, Excel, SQL, and Web sources using Pandas, and exported into different formats. The imported data was analyzed and visualized successfully.